

DB Optimization Impact Map

This document records the storage-optimization changes and the app areas affected by each table or column.

Changes Applied

Table	Removed / redirected	Replacement dependency	Affected functionality
library_books	Dropped legacy table	librarian_books	Librarian book list, issue book, return book, reminders
fee_records	Dropped legacy table	student_fee_records with fee_categories and accountant_notes	Finance dashboard, fee status updates, fee reminders
librarians	Dropped unused table	profiles.role = 'Librarian'	Librarian login/permissions
assignment_submissions.student_email	Dropped duplicate email	assignment_submissions.student_id -> profiles.email	Student assignment submission status, teacher submission export
attendance_records.student_name	Dropped duplicate name	attendance_records.student_id -> students.name	Manual attendance, AI attendance save, attendance reports
student_marks.student_name	Dropped duplicate name	student_marks.student_id -> students.name	Marks entry, student marks, teacher performance, admin marks dashboard
student_marks.class_name	Dropped duplicate class name	student_marks.section_id -> sections.name	Marks filtering, marks permissions, class exam marks
complaints.student_name	Dropped duplicate name	complaints.student_id -> students/profile name through complaint_details	Student complaint list, teacher/governing complaint inbox
complaints.class_name	Dropped duplicate class name	students.section_id -> sections.name through complaint_details	Complaint routing display and inbox filtering
complaints.section	Dropped duplicate section text	Derived from sections.name through complaint_details	Complaint display cards
complaints.division	Dropped duplicate division	Derived from students.gender through complaint_details	Complaint division display
timetable_entries.teacher_profile_id	Dropped duplicate profile reference	timetable_entries.teacher_id -> teachers.profile_id	Teacher timetable filtering and timetable RLS

Query Redirects

Functionality	Old query source	New query source
Assignment submission email	assignment_submissions.student_email	nested profiles.email via student_id
Finance fee loading	student_fee_records with fallback to fee_records	student_fee_records only
Complaint display	complaints denormalized columns	complaint_details view
Complaint creation	submit_complaint inserted denormalized student/class fields	submit_complaint inserts only IDs and complaint body

Functionality	Old query source	New query source
Marks by class	student_marks.class_name	student_marks.section_id joined to sections.name
Marks student labels	student_marks.student_name	students.name
Timetable teacher filter	timetable_entries.teacher_profile_id	teachers.profile_id joined through teacher_id
Attendance writes	writes student_name into attendance row	writes student_id only

Tables Kept Intentionally

Table / field	Why it remains
students.grade_id, students.class_id	Finance and older grade/class architecture still use these in DB functions and migration logic.
student_fee_records.grade_id, student_fee_records.class_id	Needed for fee generation/filtering and indexed finance queries.
section_teacher_assignments.teacher_profile_id	Still used by recipient routing, auth context, leave routing, and several RLS helper functions. It can be removed later, but it requires a wider rewrite of leave, auth, routing, and teacher-scope policies.
leave_requests.student_name, leave_requests.class_name, leave_requests.teacher_profile_id	Leave request RPCs and staff inbox policies still depend on them. A separate leave-module refactor is safer.
study_materials.class_name, study_materials.teacher_profile_id	Current study-material scoping policies and frontend filters use both.
assignments.class_name	Assignment visibility is class-name based and used by policies and frontend filtering.
librarian_books.status	Derived from copy counts, but selected by UI and maintained by trigger for fast display.

Verification Needed After Applying Migration

1. Student login: open assignments and submit a Drive link.
2. Teacher login: export assignment submissions and enter marks.
3. Admin login: open institution marks dashboard and filter by class.
4. Student login: submit a complaint.
5. Teacher/governing login: open complaint inbox and resolve a complaint.
6. Teacher/student login: open timetable.
7. Accountant login: open finance dashboard, update fee status, and send reminders.

8. Librarian login: add/edit/delete books, issue a book, return it, and send reminders.